

Aim

To simulate a 16-channel Massive MIMO receiver using multi-channel decimation with a sampling rate of 122.88 MHz and a decimation factor of 4, and to analyze the computational reduction and channel spectra before and after decimation.

Tools Used

- **MATLAB** / GNU Octave
- Signal Processing Toolbox
- MATLAB Plot/FFT functions

MATLAB Program:

```
clc;
clear;
close all;

%% Parameters
Nch = 16;          % Number of channels
Fs = 122.88e6;     % Sampling frequency = 122.88 MHz
D = 4;            % Decimation factor
Fs_new = Fs / D;   % New sampling frequency = 30.72 MHz

N = 4096;          % Number of samples
t = (0:N-1)/Fs;

%% Step 1: Generate Multi-Channel Signals
signals = zeros(Nch, N);

for ch = 1:Nch

    % Different frequencies for each channel
    f1 = 5e6 + (ch-1)*0.5e6;
    f2 = 12e6 + (ch-1)*0.25e6;

    signals(ch,:) = sin(2*pi*f1*t) + ...
        0.5*sin(2*pi*f2*t);
end

%% Step 2: Design Common Decimation Filter

Fpass = 12e6;
Fstop = 15e6;

filter_order = 100;

b = firpm(filter_order, ...
    [0 Fpass Fstop Fs/2]/(Fs/2), ...
    [1 1 0 0]);

%% Step 3: Perform Decimation

decimated_signals = zeros(Nch, ceil(N/D));

for ch = 1:Nch

    % Filtering
    filtered_signal = filter(b, 1, signals(ch,:));
```

```

    % Downsampling by factor 4
    decimated_signals(ch,:) = ...
        filtered_signal(1:D:end);
end

%% Step 4: Computational Reduction

input_samples = N * Nch;
output_samples = size(decimated_signals,2) * Nch;

reduction = ...
    (1 - output_samples/input_samples) * 100;

fprintf('-----\n');
fprintf('Multi-Channel Decimation for Massive MIMO\n');
fprintf('-----\n');

fprintf('Number of Channels    = %d\n', Nch);
fprintf('Original Sampling Rate = %.2f MHz\n', Fs/1e6);
fprintf('Decimation Factor      = %d\n', D);
fprintf('New Sampling Rate      = %.2f MHz\n', Fs_new/1e6);

fprintf('Input Samples          = %d\n', input_samples);
fprintf('Output Samples         = %d\n', output_samples);

fprintf('Computational Reduction = %.2f %%\n', reduction);

%% Step 5: Plot Original Signal

figure;

plot(t(1:500)*1e6, signals(1,1:500));

xlabel('Time (\mus)');
ylabel('Amplitude');

title('Original Signal - Channel 1');

grid on;

%% Step 6: Plot Decimated Signal

t_new = (0:size(decimated_signals,2)-1)/Fs_new;

figure;

plot(t_new(1:125)*1e6, ...
    decimated_signals(1,1:125));

xlabel('Time (\mus)');
ylabel('Amplitude');

title('Decimated Signal - Channel 1');

grid on;

```

%% Step 7: Compare Channel Spectra

figure;

for ch = 1:Nch

subplot(4,4,ch);

x = decimated_signals(ch,:);

L = length(x);

Y = fftshift(fft(x));

P = abs(Y/L);

f = (-L/2:L/2-1)*(Fs_new/L);

plot(f/1e6, P);

xlabel('Frequency (MHz)');

ylabel('Magnitude');

title(['Channel ', num2str(ch)]);

grid on;

end

sgtitle('Spectra of 16 Decimated MIMO Channels');

%% Step 8: Before and After Spectrum Comparison

figure;

% Original spectrum

subplot(2,1,1);

x = signals(1,:);

L = length(x);

Y = fftshift(fft(x));

P = abs(Y/L);

f = (-L/2:L/2-1)*(Fs/L);

plot(f/1e6, P);

xlabel('Frequency (MHz)');

ylabel('Normalized Magnitude');

title('Spectrum Before Decimation - Channel 1');

xlim([-30 30]);

grid on;

```

% Decimated spectrum
subplot(2,1,2);

x = decimated_signals(1,:);

L = length(x);

Y = fftshift(fft(x));

P = abs(Y/L);

f = (-L/2:L/2-1)*(Fs_new/L);

plot(f/1e6, P);

xlabel('Frequency (MHz)');
ylabel('Normalized Magnitude');

title('Spectrum After Decimation - Channel 1');

xlim([-15.36 15.36]);

grid on;

sgtitle('Multi-Channel Decimation for Massive MIMO');

%% Final Output

fprintf('\nDecimation completed successfully.\n');
fprintf('Sampling rate reduced from %.2f MHz to %.2f MHz.\n', ...
        Fs/1e6, Fs_new/1e6);

fprintf('Sample-rate related reduction = %.2f %%. \n', reduction);

```

Expected Output:

So, after decimation by a factor of 4, the sampling frequency becomes 30.72 MHz.

Output

1. The system generates 16 independent MIMO channel signals at a sampling rate of 122.88 MHz.
2. A common FIR low-pass decimation filter is designed and applied to all 16 channels.
3. The signals are downsampled by a factor of 4, reducing the number of samples to one-fourth of the original samples.
4. The sampling frequency is reduced from 122.88 MHz to 30.72 MHz, while preserving the desired signal components.
5. The spectra of all 16 channels are analyzed before and after decimation.
6. The number of processed samples is reduced from 65,536 to 16,384, resulting in approximately 75% computational/sample-rate reduction.

Result

Multi-channel decimation by a factor of 4 was successfully simulated for a 16-channel Massive MIMO receiver. The sampling rate was reduced from 122.88 MHz to 30.72 MHz, and the common FIR low-pass filter effectively suppressed unwanted high-frequency components before

downsampling. This resulted in approximately 75% reduction in the number of samples, reducing computational and memory requirements while preserving the desired channel signals.

GRAPH:

